

A SMART PILLBOX For Timely Medication

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As families grow busier, maintaining medication schedules for seniors, home-bound patients, and children who may not fully understand medication routines can become increasingly difficult, especially when medicines must be taken at irregular intervals of two to four days.

This pillbox offers a low-cost, fully electronic, circuit-based solution, requiring minimal human intervention. The user only needs to fill the weekly dosage and ensure uninterrupted power supply.

The system provides medication reminders through both visual and audio alerts. As continuous audio signals may irritate patients, special care has been taken in the design. To prevent false triggering in the audio section (using the BT66/UM66 IC), optical isolation has been implemented through an optocoupler IC.

Each day, the reminder circuit activates a corresponding LED for about ten seconds to indicate the specific day of the week. In addition, an audio alert is generated for a preset duration (again, ten seconds in the present design) at every 24-hour interval.

After the ten-second alerts each day, the system resets itself to initiate the next 24-hour cycle. For demonstration purposes, the timing interval can be reduced to ten seconds (from 24 hours) and the alert duration to one second (from ten seconds), making it more convenient for video recording and testing.

The proposed setup functions as a combined LED and buzzer alert system. Fig. 1 shows the author's prototype of the complete setup, while Fig. 2 shows the system's

block diagram. To build this system, the required components are

listed under Table 1 for the Bill of Materials.

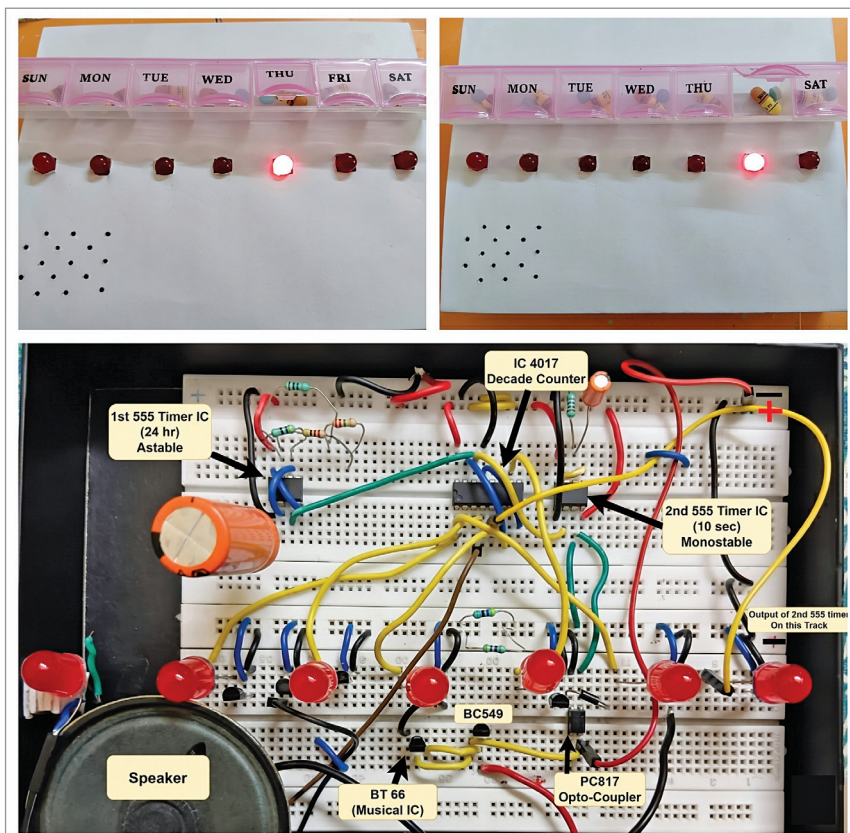


Fig. 1: Authors' prototype

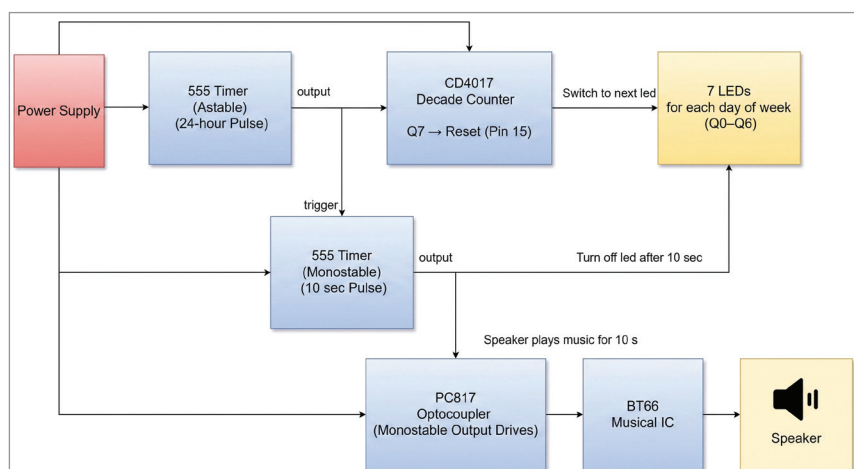


Fig. 2: Block diagram of the system